

Your infrastructure is green. Your agent is failing. You cannot see it yet.

Traditional SLOs measure infrastructure health: latency, availability, error rate, throughput. For request-response APIs and microservices those dimensions cover the primary infrastructure failure modes.

- **For an agentic system** the infrastructure can be completely healthy while the system is failing at the task level.
- **A 200 OK response** tells you the infrastructure responded. It tells you nothing about whether the agent completed the job correctly.
- **Five SLO dimensions** covering the agentic AI Performance and Scalability.



Every step passed. The end-to-end workflow still failed.

8 steps. Each is an LLM call, tool call, retrieval or sub-agent invocation. Each succeeds 95% of the time. But failures compound across the chain.

95% per step

66 of 100 workflows complete

34 fail silently

99% per step

92 of 100 workflows complete

8 fail silently

- **Moving from 95% to 99% per step** is not a 4% improvement. It is 26 more customers per 100 getting the outcome they asked for.
- **No alert fired.** No error thrown. Every agent step reported success.
- **Teams measuring SLOs** at the span level will not see this. That is the measurement gap.



SLO Dimension 1: Task Completion Rate

Every layer logged success. The workflow never fully completed.

500 cases per hour. Every layer reports success. Latency within SLA. Error rate zero.

- **160 out of 500 cases** returned a response but never completed the workflow correctly.
- **Sub-agents returned** partial outputs the orchestrator accepted and passed forward.
- **No timeout. No error.** Every agent step reported success.

A response delivered and a workflow completing correctly are not the same thing.

How to instrument: Track two numbers separately: traces that finished and traces that finished correctly. The gap between them is your task completion rate SLO.

SLO target: X% of traces reaching verified completion within Y steps and Z token budget. Define per workflow type, not globally.



SLO Dimension 2: Step Efficiency

Task completed. Four times more steps than the baseline.

An agent taking 18 steps for a task with a 4-step baseline is performing unpredictably. Extra steps cost tokens, add latency and signal something is wrong.

- **A research agent** with a 5-step baseline starts averaging 12 steps after a prompt update.
- **Latency SLO passes.** Error rate zero. Step efficiency SLO catches the regression immediately.
- **Without an SLO on step count** the degradation is invisible until the token bill arrives.

The right number of steps is the baseline. Anything materially above it is a signal.

How to instrument: Track average steps per task per workflow type. Establish baseline over two weeks of stable production traffic.

SLO target: Alert when average steps exceed 150% of baseline for structured workflows. Adjust threshold per task type. Never apply a global threshold.



SLO Dimension 3: LLM Capacity and Throttling

Agent failed mid-task. Provider hit rate limit. This is a capacity planning failure.

When an LLM provider throttles mid-workflow the agent stalls or fails the task. Standard API monitoring shows elevated latency and nothing else.

- **A single agent trace** can trigger 10 to 50 LLM calls depending on workflow complexity.
- **Capacity planning** built on single-call assumptions consistently underestimates trace-level token demand.
- **Throttling events** mid-workflow are invisible unless instrumented at the span level inside the trace.

Plan token budget, capacity at the trace level, not the request level.

How to instrument: Aggregate throttling events per trace not per request. One trace with 3 throttling events is more serious than 3 requests each with 1.

SLO target: Zero throttling events for business-critical workflows. Define tuned Token throttling per minute (TPM) and tolerance per workflow type based on business impact.



SLO Dimension 4: Average Tokens Per Trace

Token usage per trace spiked after your last agent change.

Your leading indicator SLO. Catches context accumulation, unexpected loops and retrieval drift before they become cost problems.

- **A billing agent** with an 8,000-token trace starts averaging 24,000 tokens after a new tool was added.
- **Per-call metrics look fine.** Cost looks fine. The trace-level SLO catches the drift three weeks before the monthly bill.
- **Review immediately** after any agent change, prompt update or new tool addition.

Token drift is a leading indicator. Cost impact is the lagging consequence.

How to instrument: Sum input and output tokens across all generation spans within a single trace. Track the daily average in Langfuse or your observability platform.

SLO target: Alert when 7-day rolling average exceeds the 4-week baseline by more than 2 standard deviations. Adjust sensitivity based on your workflow's natural token variance.



SLO Dimension 5: Cost Per Successful Task

Cost per request is incomplete. Measure cost per successful task.

100 tasks attempted. 70 completed correctly. \$50 spent in total.

\$50 / 100 attempts = \$0.50 per attempt **Cost per attempt**

\$50 / 70 completions = \$0.71 per successful task **Cost per outcome**

- **The \$0.21 difference** is the cost of 30 failed runs included in the numerator.
- **Each retry** adds inference cost without adding a successful completion.
- **A rising retry rate** inflates this metric faster than total spend alone suggests.

How to instrument: Use trace-level tagging to attribute costs per task type. Do not aggregate across task types with different complexity profiles.

SLO target: Cost per successful task within acceptable variance of baseline. Review whenever retry rate increases.



Five SLOs to start. One sprint per dimension.

- 1 Task completion rate: verified completion at trace level, not span level.
- 2 Step efficiency: acceptable steps per task relative to baseline.
- 3 LLM capacity and throttling: rate limiting impact at trace level.
- 4 Average tokens per trace: token drift after agent changes.
- 5 Cost per successful task: true cost including retries and failures.

Sixth dimension to add: output consistency. Same input, same outcome within acceptable variance. Non-determinism without measurement is a reliability gap.

Agentic SLOs are an engineering discipline. Define the budget. Measure against it. Alert when it is breached.

